

Dynamic Programming theory -- key ideas -- not formal

$$(RP): \quad V(k) = \max_{k' \in T(k)} \{ F(k, k') + \beta V(k') \}$$

Recursive problem

Does (RP) have a unique solution?

Observe the structure of it:

RHS: transforms $V(k)$ into something else by applying the max function

LHS: just $V(k)$ by itself

Denote the RHS of (RP) by "operator"

$$TV(k) = \max_{k' \in T(k)} \{ F(k, k') + \beta V(k') \}$$

$\Rightarrow$ (RP) becomes $\boxed{V = TV}$,

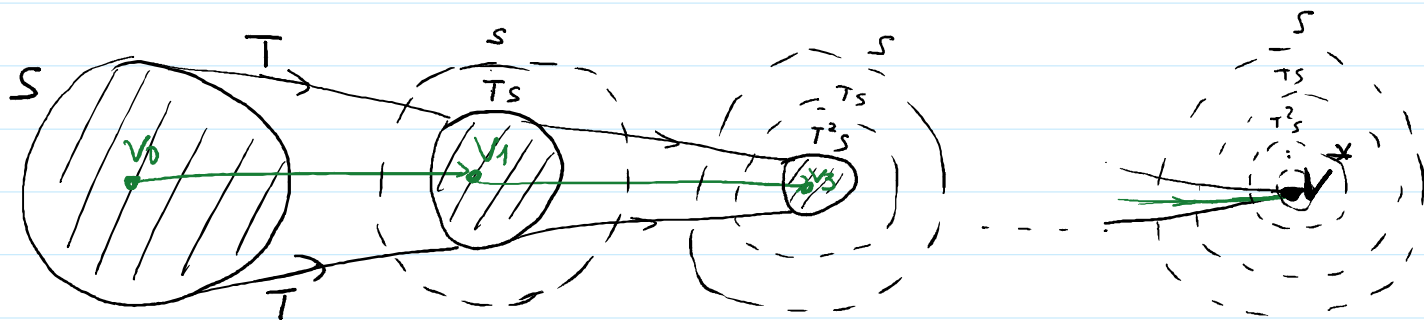
i.e. V is the "fixed point" of operator T

Thus (RP) has a unique solution if and only if operator T has a unique fixed point.

Under some conditions, T is a "contraction mapping"
 $\Rightarrow \exists!$ (there exists unique) fixed point.

Graphical intuition:

S - space of possible functions V



- If T is a "contraction" ($\approx$ maps a space into its interior)
 $\Rightarrow \exists!$ fixed point V^* that maps into itself:
$$V^* = TV^*$$

- This fixed point can be found by taking any $V_0 \in S$, and applying T to it infinitely many times.

Next, let's develop a rigorous argument supporting this "intuition".

Df (Metric space)

A metric space (S, d) is a set S and a metric (distance function) $d: S^2 \rightarrow \mathbb{R}_+$ such that:

- (i) strict positivity: $\forall x, y \in S: d(x, y) \geq 0$, " $=$ " iff $x = y$
- (ii) symmetry: $\forall x, y \in S: d(x, y) = d(y, x)$
- (iii) triangle inequality: $\forall x, y, z \in S: d(x, y) \leq d(x, z) + d(z, y)$

Note: (i)-(iii) are the basic properties of Euclidean distance

Example: $d(x, y) = \sup_{t \in \mathbb{R}} |x(t) - y(t)|$

Df (Cauchy sequence) (S, d)

A sequence $\{x_n\}_{n=0}^{\infty}$ in a metric space (S, d) is a Cauchy sequence iff $\forall \epsilon > 0 \exists N \in \mathbb{R}_+ : \forall n, m > N \quad d(x_n, x_m) < \epsilon$

Df (a Complete Metric Space)

A metric space is complete if each Cauchy sequence converges to a limit in the metric space

Df (Contraction) $T: S \rightarrow S$

An operator T is a contraction iff $\exists \gamma \in [0, 1) : \forall x_1, x_2 \in S$
 $d(T(x_1), T(x_2)) \leq \gamma d(x_1, x_2)$.
Here γ is called "the modulus".

Theorem: (Contraction Mapping)

Suppose that (S, d) is a complete metric space and $T: S \rightarrow S$ is a contraction of modulus γ .

Then:

- $\exists! \hat{x} \in S : T(\hat{x}) = \hat{x}$ (unique fixed point)
- $\forall x_0 \in S \quad \{T^n(x_0)\}_{n=1}^{\infty}$ converges to $\hat{x}$

Contraction Mapping Theorem

Stated last time:

Theorem: (Contraction Mapping)

Suppose that (S, d) is a complete metric space and $T: S \rightarrow S$ is a contraction of modulus δ .

Then:

- $\exists! \hat{x} \in S : T(\hat{x}) = \hat{x}$ (unique fixed point)
- $\forall x_0 \in S \quad \{T^n(x_0)\}_{n=1}^{\infty}$ converges to $\hat{x}$

Proof:

Steps: ① Show that $\exists \lim_{n \rightarrow \infty} T^n(x_0) = \hat{x}$ (i.e. that $\{x_n\}_{n \geq 1}$ is a Cauchy sequence)

② Show that $\hat{x} = \lim_{n \rightarrow \infty} x_n$ is a fixed point ($\Rightarrow \exists$ fixed point)

③ Show that the fixed point is unique

① Need to show that $\forall \varepsilon \exists N: \forall n, m > N: d(T^n(x_0), T^m(x_0)) < \varepsilon$
Let $m < n$

$$d(T^n(x_0), T^m(x_0)) = d(T^m(T^{n-m}(x_0)), T^m(x_0)) \leq (\text{T-contract})$$

$$\leq \delta^m d(T^{n-m}(x_0), x_0)$$

$$= \delta^m d(T^{n-m-1}(x_1), x_0) \leq |\text{triangle}| \leq$$

$$\leq \delta^m \cdot [d(T^{n-m-1}(x_1), T^{n-m-1}(x_0)) + d(T^{n-m-1}(x_0), x_0)]$$

(T-contract.)

$$\leq \delta^m \cdot [\delta^{n-m-1} d(x_1, x_0) + d(T^{n-m-2}(x_1), x_0)] \leq$$

(triangle)

$$\leq \delta^m \cdot [\delta^{n-m-1} d(x_1, x_0) + d(T^{n-m-2}(x_1), T^{n-m-2}(x_0)) + d(T^{n-m-2}(x_0), x_0)] \leq$$

$$\leq (\text{contract}) \leq \delta^m \cdot [\delta^{n-m-1} d(x_1, x_0) + \delta^{n-m-2} d(x_1, x_0) + d(T^{n-m-3}(x_1), x_0)] \leq \dots$$

apply the same steps

$$\leq \delta^m \cdot [\delta^{n-m-1} + \delta^{n-m-2} + \delta^{n-m-3} + \dots + \delta^1 + 1] \cdot d(x_1, x_0)$$

$$= \delta^m \cdot \frac{1-\delta^{n-m}}{1-\delta} d(x_1, x_0) \leq \Sigma \text{ for } m \text{ sufficiently large.}$$

$\Rightarrow \{T^n(x_0)\}$ is a Cauchy sequence

② Let $\hat{x} = \lim_{n \rightarrow \infty} T^n(x_0) \in S$. Need to show that $d(\hat{x}, T(\hat{x})) = 0$
(by completeness of (S, d))

$$\begin{aligned} d(\hat{x}, T(\hat{x})) &\leq d(\hat{x}, T^n(x_0)) + d(T^n(x_0), T(\hat{x})) \\ &\quad \uparrow \\ &\quad \text{triangle} \\ &\leq \underbrace{d(\hat{x}, T^n(x_0))}_{\rightarrow 0, n \rightarrow \infty} + \delta \underbrace{d(T^{n-1}(x_0), \hat{x})}_{\rightarrow 0, n \rightarrow \infty} \\ &\quad \text{T-contraction.} \rightarrow \end{aligned}$$

$$\Rightarrow d(\hat{x}, T(\hat{x})) = 0$$

③ $\hat{x}$ is a unique fixed point
(i.e. $T^n(x_0)$ converges to the same $\hat{x}$ from any x_0)

By contradiction:

suppose there is another fixed point $\hat{y}$

$$\begin{aligned} \Rightarrow \underline{d(\hat{x}, \hat{y})} &\leq \underbrace{d(\hat{x}, T(\hat{x}))}_{=0} + d(T(\hat{x}), T(\hat{y})) + \underbrace{d(T(\hat{y}), \hat{y})}_{=0} \\ &\leq (\text{contraction}) \leq \delta \underline{d(\hat{x}, \hat{y})} \end{aligned}$$

$$\Rightarrow d(\hat{x}, \hat{y}) = 0$$



Blackwell Sufficient Condition

Thus, we need to check that the operator T (in the RHS of the Bellman Equation) is a contraction.

Namely, if V, W are two functions then

$$d^v(T(v), T(w)) \leq \gamma d(v, w) \quad \text{for some } \gamma < 1$$

So we need to define metric d and, possibly, impose a restriction on the set of functions.

Let's use the sup norm

$$\|V\| = \sup_{k \in X} |V(k)| \quad \text{where } X \text{ is the set on which } f\text{-n } V \text{ is defined}$$

$$d(v, w) = \|v - w\| = \sup_{k \in X} |v(k) - w(k)|$$

For this metric to be well-defined, $V \in W$ must be bounded.

The following result establishes sufficient conditions under which operator T in the metric space of (bounded functions, sup norm) is a contraction:

Theorem: Blackwell's Sufficient Condition

Let $B(X)$ be a set of bounded functions on X , and suppose that $T: B \rightarrow B$ satisfies

1) Monotonicity: $\forall V, W \in \mathcal{B}(X)$: $V \leq W \Rightarrow T(V) \leq T(W)$
 pointwise

2) Discounting : $\exists \delta \in [0,1) : \forall c \in \mathbb{R}_+ \quad \forall v \in B(x)$

$$T(\underbrace{v+c}_{\text{constant}}) \leq Tv + \delta c$$

Then T is a contraction of modulus γ w.r.t. sup. metrics

Proof: Need to show that

$$\sup_{k \in X} |(TV)(k) - (TW)(k)| \leq \gamma \sup_{k \in X} |V(k) - W(k)|$$

$$\begin{aligned} \forall k: \quad V(k) &= W(k) + [V(k) - W(k)] \\ &\leq W(k) + \sup_k |V(k) - W(k)| = W(k) + d(V, W) \end{aligned}$$

$$\begin{aligned} \text{Monotonicity} \Rightarrow TV &\leq T(W + \underbrace{d(V, W)}_{\text{constant}}) \leq \text{discounting} \\ &\leq TW + \gamma \cdot d(V, W) \end{aligned}$$

$$\Rightarrow TV - TW \leq \gamma \cdot d(V, W)$$

$$\text{Likewise, } W(k) = V(k) + [W(k) - V(k)]$$

--- same arguments ---

$$\Rightarrow TW - TV \leq \gamma \cdot d(W, V) = \gamma \cdot d(V, W) \quad (\text{symmetry of } d(V, W))$$

$$\Rightarrow -\gamma \cdot d(V, W) \leq TV - TW \leq \gamma \cdot d(V, W)$$

$$\Rightarrow \sup_k |(TV)(k) - (TW)(k)| \leq \gamma \cdot d(V, W) = \gamma \cdot \sup_k |V(k) - W(k)|$$

